Duke UNIVERSITY | ELECTRICAL & COMPUTER ENGINEERING

Undergraduate Program Handbook

For Students Matriculating Summer 2025 or Before

vF25.2 2025-2026 Academic Year

The handbook is updated every year.
Students and faculty should always use the latest version of the handbook.

Highlights of Important Changes from F24 Version

- ***<u>The Trinity College of Arts and Sciences has approved a new curriculum, which includes new curricular codes – given that, ECE has a handbook specifically for those students. Students who matriculate in Fall 2025 or later should use that new handbook, not this one. Students who matriculated before Fall of 2025 should use this handbook.</u>***
- The Physics Department *strongly recommends* MATH 219 as a pre-requisite or co-requisite for PHYSICS 152.
- The ECE Department has approved a Transcripted Concentration in Software Engineering and a Minor in Software Engineering
- Previously, the terms "ECE Concentration Elective" and "ECE Concentration Course" were used to describe four courses that needed to be selected from specific lists among the five curricular groups in ECE. Now that ECE has two transcripted concentrations, the terms "ECE Approved Curricular Area Elective" and "ECE Approved Curricular Area Course" will be used. These terms may show up in their shortened forms of "ECE Area Elective" and "ECE Area Course."
- Several new ECE Area Electives have been approved:
    - Computer Engineering and Digital Systems:
        - *ECE 495 Software Development and Engineering*
        - *ECE 557 Computer Architecture and Hardware Acceleration*
        - *ECE 590 Cross-Platform Mobile Application Programming*
        - *ECE 564 Mobile Application Programming*
        - *ECE 657 Usable Security and Privacy*
        - *ECE 653 Human-Centered Computing*
    - Engineering Physics
        - *ECE 522 Quantum Engineering with Atoms*
    - Solid-State Devices and Integrated Circuits
        - *ECE 431 Power Electronic Circuits for Energy Conversion*
        - *ECE 516 (c/l ME 516) Thin-Film Photovoltaic Technology*
    - Signal Processing, Communications, and Controls
        - *ECE 590 Neural Network Based Large Language Models*
        - *ECE 661 Computer Engineering Machine Learning and Deep Neural Networks*
- The Computer Engineering & Digital Systems (CEDS) area no longer has a foundation course. Students wishing to take multiple courses in CEDS are no longer required to take ECE 350L for more than one CEDS course to count as an area course.
- *ECE 590 Neural Network Based Large Language Models* has been approved as an Option 3/4/5 course for the Transcripted Concentration and Minor in Machine Learning & AI.

- ECE 590 Neural Network Based Large Language Models has been approved as an Option 3/4/5 course for the Transcripted Concentration and Minor in Machine Learning & AI.
- ECE 351 Software Engineering has been given a permanent number. It will be included as an alternative prerequisite for any ECE course currently listing COMPSCI 307D or COMPSCI 308 as prerequisites.
- ECE 522 Quantum Engineering with Atoms has been given a permanent number.
- PHYSICS 385 is now cross-listed as ECE 332.
- The following courses have been removed from the system:
    - *ECE 486 Wireless Communication Systems*
    - *ECE 525 Semiconductor Physics*
    - *ECE 562 Energy-Efficient Computer Systems*
    - *ECE 578 Inverse Problems in Electromagnetics and Acoustics*
    - *ECE 584 Acoustics and Hearing*
    - *ECE 611 Nanoscale and Molecular Scale Computing*
    - *ECE 676 Lens Design*
- The following courses, while still in the system, are considered dormant and will not be included in this handbook:
    - *ECE 488 Digital Image and Multidimensional Processing*
    - *ECE 489 Advanced Robot System Design*
    - *ECE 496 Wireless Technology and Wave Propagation*
    - *ECE 561 Datacenter Architecture*
    - *ECE 563 Cloud Computing*
    - *ECE 573 Optical Communication Systems*
    - *ECE 577 Computational Electromagnetics*
    - *ECE 582 Digital Signal Processing*
    - *ECE 631 Analog and RF Integrated Circuit Design, Fab and Test*
    - *ECE 662 Machine Learning Acceleration and Neuromorphic Computing*
    - *ECE 681 Pattern Classification and Recognition Technology*
    - *ECE 683 Digital Communication Systems*
    - *ECE 686 Adaptive Filters*

Version Changes:

- v1.1 10NOV2025: Fixed Table of Contents

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Introduction

This handbook provides an overview of the undergraduate program in Electrical and Computer Engineering (ECE) at Duke University. It covers the program, mission, educational objectives, major requirements, second major options, minor requirements, and research opportunities.

Duke's Department of Electrical and Computer engineering offers an ABET accredited Bachelor of Science in Engineering emphasizing analytical thinking, problem solving, and technology, while fostering the development of ethical leadership, communication, and teamwork skills. The Duke program provides more opportunities and is less restrictive than many other ECE programs. Students use this flexibility to expand their horizons or to explore a personal interest more deeply. The student population is diverse, both geographically and culturally, and is a cross-section of the very best students in the nation and from around the world.

For more information about the ECE undergraduate program contact:

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Mission

Electrical and computer engineering is a broadly based engineering discipline dealing with the processing, control, and transmission of information and energy by making extensive use of electrical and electromagnetic phenomena, systems theory, and computational hardware and software. The Department also encourages students and faculty to develop synergies with disciplines outside of engineering, such as medicine and the life sciences. Electrical and computer engineers can complete a second major or a minor in many fields such as biomedical engineering, computer science, physics, mathematics, and economics. Additional interests such as pre-medicine, pre-law, business, other engineering disciplines, art, music, psychology, and social sciences can be accommodated.

The mission of the Department of Electrical and Computer Engineering is to facilitate the development of well rounded, educated, productive, and ethical individuals who are well versed in technology and in social, political, and environmental issues. Our goals are to develop within each student a robust repertoire of professional skills, to provide each with avenues for exploring diverse interests, and to launch each successfully into one of a variety of careers offering lifelong learning, service, and leadership within their own local, national, and global communities. To achieve our mission, the department puts forth the following educational objectives for the extremely capable students entering the ECE program.

Program Educational Objectives

Our goal is to graduate electrical and computer engineers who embody excellence in a broad sense. We expect our graduates to advance within industry positions or in graduate study, or to carry the attributes of an engineering education into other disciplines. The electrical and computer engineering program of study must include mathematics and basic sciences, fundamentals and applications in several engineering sciences, and team-based experience in the process of design, where theory is applied in the context of real needs and limitations, and where judgment must be exercised. Our electrical and computer engineering graduates should be able to think critically when solving problems and managing tasks and communicate effectively in multidisciplinary professional environments. To be a responsible member of the engineering profession, each graduate must be aware of social, ethical, environmental and economic factors and constraints on engineering activity, and must understand the importance of these matters in a global context. We aspire to have our graduates exhibit intellectual depth and creativity, uphold high ethical standards, and show a commitment to the betterment of society through service and professional work.

The specific Program Educational Objectives that we look for in our graduates are that they:

- **Advance professionally** in their chosen field
- **Contribute to their professional community and to society**
- Engage in **lifelong learning** in professional and personal endeavors

Student Outcomes and Program Criteria

Our students will have the following capabilities upon completion of their degrees:

1. an ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. an ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. an ability to communicate effectively with a range of audiences
4. an ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. an ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. an ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. an ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Pratt School of Engineering and ECE Major Requirements

As a program accredited by the Engineering Accreditation Commission of ABET (ABET website), the ECE curricula must satisfy minimum requirements in mathematics, sciences, and engineering. In addition, the Pratt School of Engineering has requirements for all engineering students. To meet these constraints, the Department of Electrical and Computer Engineering has developed specific requirements for undergraduate students. Samples of curricula for students with different ECE-related major choices are provided in Appendix A. In summary, ECE students need to take the following courses:

Five Social Sciences and Humanities (SSH)

Students matriculating before Fall of 2025 must take five (5) humanities and social science courses from approved departments and programs (see the "Degree Requirements, Engineering" section at https://pratt.duke.edu/life/resources/policies/#policies for a complete list) subject to the following requirements:

- **Social Science**: At least one course must be a Social Science (SS).
- **Breadth**: At least two of the following three areas of knowledge must be covered: Arts, Literature, and Performance (ALP), Civilizations (CZ), and Foreign Language (FL).
- **Depth**: At least two courses must be taken from the same department subject code, with at least one of the two courses at the 200-level or higher.

and subject to the following restrictions:

- At most, 2 of the 5 SSH course credits can be met by any combination of two from:
  - AP credit / IPC credit
  - Duke-offered courses that are only offered on an S/U grading basis.
- Courses utilized to meet the SSH requirement can contribute only a single area of knowledge code even if the course possesses multiple codes.
- AP credit cannot be used to meet the "200-level or higher"-component of the depth requirement
- Skill courses cannot be used
- Courses must be from, or cross-listed with, an approved department or program. See the "Degree Requirements, Engineering" section at https://pratt.duke.edu/life/resources/policies/#policies for a brief list of exceptions to this restriction.

Two Physics Courses

Engineering students must take at least one Physics course post-matriculation. Students with no AP credit will take the following:

- Physics 151L + Physics 152L

Students with a 4 or 5 on the AP Physics C exam(s) earn Physics 25-*Mechanics*; and/or Physics 26-*Electricity & Magnetism*. The following options are available:

- No AP credit or AP<4: must take Physics 151L and Physics 152L
- Physics 25 (AP>=4 on Physics C Mech) *only* + Physics 152L
- Physics 26 (AP>=4 on Physics C E&M) *only* + Physics 151L
- Physics 25 *and* Physics 26 + A Physics course at Duke (see below)

If you have AP credit for Physics 25 and 26, choose one of the following options as your Physics course post-matriculation:

- If you are interested in a second major or a minor in physics, it is suggested that you follow the recommendations for potential physics majors and minors (https://physics.duke.edu/undergraduate/course-selection/intro-placement).
- If you are not planning a second major or a minor in physics, and you have credit for multivariable calculus, it is suggested that you take Phys 264L: Optics & Modern Physics.
- If you do not have credit for multivariable calculus and are not planning a second major or minor in physics, we recommend that you take one of the following:
    - Physics 152L, or
    - an Intermediate Core Physics course, selected from: Physics 361, 362D, 363, 464, or 513, or
    - a Gateway Core Physics course, selected from: Physics 305, 320L, 380, or 414

One Chemistry Course

This requirement may be satisfied by taking Chemistry 101DL. AP credit in Chemistry [20 or 21] may be used to satisfy this requirement.

Five Mathematics Courses

All ECE majors must take Math 111L (or have AP credit for Math 21), 112L or 122L (or have AP credit for Math 22), 218D-2, 219, and 353. Common questions about mathematics placement are answered on the Math Department Placement Guideline website. Transfer credits are examined on an individual basis.

Math options for the first two courses:

No AP: Math 111L **and** 112L
Math 21 AP: Math 122L in the Fall **or** Math 112L in the Spring
Math 21 AP **and** Math 22 AP: proceed to Math 218D-2

If a student is advised by the Math department to skip any courses in the Math sequence listed above, *those courses must be replaced with additional Math courses approved by the ECE DUS.*

Special Note for students with a second major in Math: Students wishing to complete a second major in Math will need to take Math 111L, 112L, 221, 222, and 356. *Additional math courses are required to complete the Math major – check with the DUS in Math for details.*

One Statistics Course

All ECE majors must take a course on Statistics/Probability. The approved options can be found in Table E.4 Approved ECE Statistics Courses.

One Undergraduate Writing Course

Writing 120 (formerly Writing 101) is required. Generally engineers will take Writing 120 in the spring of their first year.

Engineering Design and Communication

EGR 101L is required of all Pratt students in their first year. Students transferring to Pratt who did not take EGR 101L in their first year must take an additional 300-level or higher ECE course that is approved by the Director of Undergraduate Studies in lieu of EGR 101L.

Engineering Computation and Computer Programming
All ECE students must take EGR 103L (if they matriculated prior to Fall 2024) or EGR 105L (if they matriculated Fall 2024 or later) and Computer Science 201. NOTE: Students who place into Computer Science 201 (e.g., by having CS AP credit) are not required to take EGR 103L or EGR 105L. Either EGR 103L or EGR 105L or Computer Science 201 must be taken by the end of the first year.

ECE Required Courses (Core)
The following five core courses are required for all ECE majors: ECE 110L, ECE 230L, ECE 250D, ECE 270DL, ECE 280L.

ECE Required Courses (Approved Curricular Area Electives)
ECE majors must select a minimum of four upper-level courses in at least two departmental curricular areas, with no fewer than two courses in one of the selected areas. For students expecting to enter the engineering profession after graduation, a two- or three-course sequence prepares the student for professional work in that area. For all students, including those expecting to enter fields such as medicine, law, or business, these upper-level courses reinforce the broad relevance of the powerful problem-solving methodologies of engineering and illuminate enabling technologies for breathtaking applications of technology.

The four ECE Approved Curricular Area Electives are chosen from the following five curricular areas:

- Computer Engineering and Digital Systems (CEDS)
- Signal Processing, Communications, and Control Systems (SPCC)
- Solid-State Devices and Integrated Circuits (SSDIC)
- Engineering Physics (EP)
- Photonics (P)

A complete list of approved Curricular Area Electives in each area can be found in Appendix D: Approved ECE Curricular Area Elective.

ECE Required Courses (Approved Design Elective)
The undergraduate ECE Program includes an Approved ECE Design Elective. This course is normally taken during the Senior year and only after a student has completed the core courses in math and science as well as the core and some advanced courses in ECE. This "capstone design course" involves multidisciplinary teams of students who build and test custom designed systems, components or engineering processes. Students gain experience in the design, building, testing, and demonstration processes intrinsic to engineering design as practiced by engineering graduates.

Requirements for the team project include:

- a design plan incorporating engineering standards and realistic constraints
- a timeline indicating project milestones
- a written project report including an assessment of the results
- oral presentations to the class

The completed project assessment must include most of the following elements:

- cost
- environmental impact
- manufacturability
- ethics
- health and safety
- social and political impacts

Approved ECE design courses can be found in **Table E.1 Approved ECE Design Electives**.

ECE Required Courses (Free ECE Elective)

Students take elective courses to learn advanced knowledge in specific areas of electrical and computer engineering. Students are required to take at least one elective ECE course at the 300-level or above. An ECE Free Elective can be any ECE course for which you have the prerequisites and which is not a required course for your major. Up to one ECE independent study course (300-level or above) can be counted as an ECE Free Elective.

NOTE: For students with a second major, the total number of ECE Area Electives and ECE Free Electives required depends on the second major (see Appendix A).

ECE Required Courses (ECE Extension Elective)

Students take one ECE Extension Elective course that enables students to either deepen their knowledge in a specific area of ECE, or extend their ECE training in a complementary direction. An ECE Extension Elective can be any upper-level (300-level or above) course in Biology, Chemistry, Computer Science, Evolutionary Anthropology, Mathematics, Physics, Psychology & Neuroscience, or Statistical Science **that has an NS or QS code and does not also have an SS code;** or any upper-level (300-level or above) course in BME, CEE, ECE, or ME; or a course selected from a list of approved 200-level ECE Extension Elective courses (see Appendix E for approved list). However, *if both ECE 380 and ECE 555 are taken and neither is used to satisfy the ECE statistics requirement, then the ECE Extension Elective is restricted to be an "engineering topics" course at the 300-level or higher that has been approved by the ECE DUS*. An ECE independent study course (300-level or above) can be counted as an ECE Extension Elective. **Independent Study courses from other departments cannot be used to satisfy the ECE Extension Elective requirement. AP credits cannot be used to satisfy this requirement.**

Unrestricted Electives

The standard ECE curriculum (assuming no AP credit) includes four unrestricted elective courses. Only unrestricted electives can be taken on a Satisfactory/Unsatisfactory basis.

Requirements for a Concentration in Machine Learning

Students pursuing a major in ECE who are interested in Machine Learning may focus their course selection within the major by following a defined thematic pathway, successful completion of which will result in the designation of a ***Concentration in Machine Learning*** on their official transcript. ECE majors can declare this concentration by selecting this option on the Pratt Declaration of Major form.

Completion of the Concentration in Machine Learning requires a minimum of 5 courses. These requirements can be met within the general requirements of the ECE major and do not require any *additional* courses; however, the *choice* of classes has been constrained. The required courses are listed below, along with the ECE major requirement being fulfilled (in square brackets). No course may be used to fulfill more than one requirement.

Requirements (5 courses):

1. *ECE 480: Applied Probability for Statistical Learning* [ECE Conc. Elective #1 - SPC&C]
2. *ECE 580: Introduction to Machine Learning* [ECE Area Elective #2 - SPC&C]
3. One of the following [ECE Area Elective #3 – any area, including SPC&C]:
    - *ECE 588: Image & Video Processing*
    - *ECE 661: Computer Engineering Machine Learning & Deep Neural Networks*
    - *ECE 662: Machine Learning Acceleration & Neuromorphic Computing*
    - *ECE 663: Machine Learning in Adversarial Settings*
    - *ECE 682D/CS 571/Stat 561: Probabilistic Machine Learning*
    - *ECE 684: Natural Language Processing*
    - *ECE 685D: Introduction to Deep Learning*
    - *ECE 687D/CS 671D/Stat 671D: Theory & Algorithms for Machine Learning*
    - *ECE 495/496/590: Special Topic* courses on ML that are approved area electives (with DUS approval)
        - *ECE 495 Introduction to Natural Language Processing*
        - *ECE 590 Neural Network Based Large Language Models*
4. One of the following [ECE Free Elective]:
    - *Any course not already taken from Requirement #3 (above)*
    - *ECE 689/CS 676: Advanced Topics in Deep Learning*
    - *ECE 495/496/590: Special Topic* courses on Machine Learning (with DUS approval)
5. One of the following [ECE Extension Elective]:
    - *Any course not already taken from Requirements #3 and #4 (above)*
    - *ECE 495/496/590: Special Topics* courses on Machine Learning (with DUS approval)
    - *BME 590: Machine Learning in Pharmacology (F23)*
    - *COMPSCI 371: Elements of Machine Learning*
    - *COMPSCI 527: Computer Vision*
    - *MATH 412 / COMPSCI 434: Topological Data Analysis*
    - *MATH 465 / COMPSCI 445: Introduction to High Dimensional Data Analysis*
    - *ME 555: Robot Learning (F23)*
    - *STA 340: Introduction to Statistical Decision Analysis*
    - *STA 360: Bayesian Inference and Modern Statistical Methods*

Requirements for a Concentration in Software Engineering

Students pursuing a major in ECE who are interested in Software Engineering may focus their course selection within the major by following a defined thematic pathway, successful completion of which will result in the designation of a ***Concentration in Software Engineering*** on their official transcript. ECE majors can declare this concentration by selecting this option on the Pratt Declaration of Major form.

Completion of the Concentration in Software Engineering requires a minimum of 5 courses. These requirements can be met within the general requirements of the ECE major and do not require any *additional* courses; however, the *choice* of classes has been constrained. The required courses are listed below, along with the ECE major requirement being fulfilled (in square brackets). No course may be used to fulfill more than one requirement.

Requirements (5 courses):

1. *ECE 351: Software Engineering* [ECE Conc. Elective #1 - CEDS]
2. One of the following HCI/UX courses [ECE Area Elective #2 - CEDS]
    - *ECE 653/COMPSCI 653: Human-Centered Computing*
    - *ECE 657/COMPSCI 586: Usable Security and Privacy*
    - *ECE 590: Cross Platform Mobile Application Development*
    - *ECE 490/495/496/590: Special Topic* courses on HCI/UX (with DUS approval and
    - approved as a CEDS area course)
3. One of the following Systems courses [ECE Area Elective #3 – any area, including CEDS]:
    - *ECE 353/COMPSCI 310: Introduction to Operating Systems*
    - *ECE 356/COMPSCI 356: Computer Network Architecture*
    - *ECE 553/COMPSCI 553: Compiler Construction*
    - *ECE 560: Computer and Information Security*
    - *ECE 566: Enterprise Storage Architecture*
4. One of the following Software Engineering courses [ECE Free Elective]:
    - Any course not already taken from Requirement #2
    - Any course not already taken from Requirement #3
    - *ECE 458: Engineering Software for Maintainability*
    - *ECE 568: Engineering Robust Server Software*
    - *ECE 661: Computer Engineering Machine Learning and Deep Neural Nets*
    - *ECE 490/495/496/590: Special Topic* courses on Software Engineering (with DUS approval)
5. One of the following [ECE Extension Elective]:
    - Any course not already taken from Requirement #2
    - Any course not already taken from Requirement #3
    - Any course not already taken from Requirement #4
    - Other software-focused computer science classes with DUS approval

Requirements for a Minor in Electrical and Computer Engineering

A Minor in Electrical and Computer Engineering provides Duke undergraduates with a creditable exposure to ECE that complements the program of study in their first major. It is designed to provide students with a broad, fundamental foundation in ECE, coupled with the opportunity to explore advanced topics tailored to a student's specific interests.

The minor in ECE requires a minimum of five technical courses. Three courses must be drawn from the set of "core courses" required of all ECE majors and two must be upper-level ECE courses.

Core courses (3): Choose 3 from the following 5 core ECE courses[1]:

- *ECE 110L Fundamentals of ECE*[2]
- *ECE 230L Microelectronic Devices & Circuits*
- *ECE 250D Computer Architecture*
- *ECE 270DL Fields & Waves*
- *ECE 280L Signals & Systems*

[1] Students with credit for any of these courses (e.g., exact or equivalent course taken to satisfy a requirement of their major(s)) may substitute additional upper-level ECE courses. The DUS in ECE must approve such exceptions.

[2] Note that ECE 110L is a prerequisite for the other core courses.

Upper-level courses (2): Two ECE courses at or above the 300-level

- At most, one ECE Independent Study (supervised by an ECE faculty member) can be used toward satisfying this requirement.
- At most, one 300-level (or above) course cross-listed between ECE and the major department can be used toward satisfying this requirement. This course may not be double-counted toward the major.

Courses that are used to fulfill the student's major(s) may not be double-counted toward the minor. In addition, ECE courses with content substantially equivalent to courses in the student's major(s) may not be counted toward the minor

It is expected that a student pursuing a minor in ECE will satisfy all pre-requisites for each course selected for their minor program. This will typically involve completion of courses in Math, Physics and/or Computer Science, which are pre-requisites for many of the core ECE courses. See ***Appendix C: Major-specific Requirements for the Minor in ECE*** for major-specific course and pre-requisite equivalents.

Students interested in pursuing a Minor in ECE are advised to discuss their plan of study with the Director of Undergraduate Studies in ECE (Dr. Michael Gustafson, mrg@duke.edu).

Second Major, Minor, and Certificate Programs

Students can declare a second major within the Pratt School of Engineering. Opportunities also exist for students with AP credits to combine the ECE major with a second major, minor, or certificate from the Trinity College. (A certificate is similar to a minor but offered for interdisciplinary study.) To do so, the students must meet the same requirements as those for the ECE major plus the specific requirements from other departments/programs outlined in the Bulletin of Undergraduate Instruction. The additional requirements usually consist of ≥10 courses for a second major, 5 courses for a minor, and ≥ 6 courses for a certificate. Some of these courses can be double-counted towards both the ECE degree and the second major, minor or certificate in the Trinity College. For example, two courses required for the second major in economics may be counted as two of the five SSH courses required for the ECE degree. To reduce the work load for obtaining the second major, minor, or certificate in the Trinity College during the regular academic semesters, students can either take the required Trinity courses as unrestricted electives in the ECE curriculum or take them in the summer.

Appendix A provides samples of curricula for the most popular combinations of the ECE major with a second major.

Independent Study and Research Opportunities

Independent study courses are designed to allow Juniors and Seniors to work individually with a faculty member on a project or topic of mutual interest. Courses are arranged on an individual basis at the instigation of the student or faculty member. Most students who undertake Independent Study do so in their Senior year, but qualified students can undertake Independent Study after Sophomore year.

Many students find that Independent Study is a rewarding educational experience. Students who anticipate Independent Study are encouraged to complete their foundation courses in mathematics, chemistry, physics, and engineering as early as possible, so that they will have the background to address challenging engineering problems in collaboration with their faculty study mentor.

Up to two Independent Study courses [ECE 391, ECE 392, ECE 493 and 494] may be used to satisfy ECE major electives (e.g., as ECE Free Elective(s) or ECE Extension Elective). Students may take Independent Study courses as Free Electives, and all successfully completed Independent Study courses count toward the total of 34 courses required for a Duke degree.

A research experience can be achieved in a number of ways.

- Perform directed ECE research with an ECE faculty member.
- Take independent study courses with a non-ECE faculty member. In such cases, the project must be sponsored by an ECE faculty member. The sponsor is responsible for evaluating the quality of the project and ensuring ECE content as appropriate for ECE course credit.
- Apply for a Pratt Research Fellow position during your Junior year. Pratt Research Fellows perform research in an ECE faculty member's lab for three academic semesters plus a full summer term and receive two ECE Free Elective credits.

All students taking the ECE independent-study courses, *including Pratt Fellows and Graduation with Departmental Distinction candidates*, must participate in ECE Independent Study Poster

Session to present their research results and to answer questions from faculty and other students. Poster sessions are scheduled by the ECE Department late in each semester for the independent study done during that semester.

Other opportunities for integrating research into the plan of study include the Duke Smart Home Fellows program, the Grand Challenge Scholars program, and Bass Connections.

Graduation with Distinction

Students who aspire to Graduate with Departmental Distinction (GWDD) within the Department of Electrical and Computer Engineering conduct supervised research through independent study courses and present the results of individual research and study in both written and oral forms to the department's faculty. Those candidates who are judged by the department's faculty to have distinguished themselves through their paper and presentation earn Graduation with Departmental Distinction honors.

To be considered for Graduation with Departmental Distinction a student must have a 3.5 Cumulative GPA and must successfully complete in his or her Senior year a faculty supervised Independent Study project with significant Electrical and Computer Engineering accomplishment. The significance of this project must be demonstrated in a formal written report and defended in an oral presentation before a committee of faculty members.

Candidates must submit a 10-20 page written report (single spaced, 12-point Times New Roman, single column) including figures and references. The entire report must be submitted electronically to the Director of Undergraduate Studies Assistant no later than one week **before** the oral presentation begins. The project report essentially constitutes a Senior thesis.

This accomplishment is recognized when the BSE degree is awarded. If that time is different from the principal Spring commencement exercises, the student work remains in contention for the research prize for that academic year awarded each Spring Commencement.

Preparation for Medical School

Students planning to attend medical and dental schools should consult with the Office of Health Professions Advising (HPA) about course planning. The Office provides advice to students planning careers in health professions and also information on the application process, degree requirements, research and volunteer opportunities.

Planning for Study Abroad

Students interested in participating in the Global Education program need to develop an academic plan with their advisors several semesters in advance. This plan includes the courses to be taken in foreign countries and the remaining courses to be taken at Duke before and after the semester abroad. In the semester *prior* to study abroad, students submit requests for course approvals (if courses are not already in the GEO database) to the ECE DUS. The process for approval of courses in the study abroad program is the same as that used for all transfer courses. For details, see the Duke Global Education website.

ECE students may take two courses abroad related to the major. The remainder of courses taken abroad may fulfill other requirements. No more than two courses can be taken in place of required ECE departmental courses (e.g., ECE 280L). In special cases, with prior approval of the Director of Undergraduate Studies (DUS), students may take two required courses plus a technical elective required for the major (or second major). This situation arises most often when there is an opportunity to study a subject not normally offered at Duke

The most common courses for students to receive credit for taking abroad (i.e., the courses for which it is easiest to find an equivalent) include ECE 280L, some ECE Area Electives (particularly ECE 356, 381, 382L, 383, and 488), ECE Free Electives, and/or Social Studies or Humanities electives.

4 + 1 BSE/Master's Program (Five-Year Combined Bachelor/Master's Degree Program)

Completing both a Bachelor of Science in Engineering (BSE) and a Master's degree in five years is a great opportunity to advance your training. If you plan accordingly, you can complete all requirements for your BSE degree without carrying a full undergraduate course load in your senior spring. By taking two or more graduate courses in your senior spring (which are not being used to satisfy any BSE requirements), you can complete the remaining graduate courses in one year beyond your BSE.

At Duke there are three engineering master's degree options to consider:

- 4+1: BSE + Master of Engineering
- 4+1: BSE + Master of Science
- 4+1: BSE + Master of Engineering Management

More information about all three 4+1 programs is provided at the 4+1 webpage and a comparison of the graduate degrees can be found on the MS and Ph.D. degree website. In all cases, you want to work with your Director of Undergraduate Studies and the Director of Graduate Studies to develop your academic plan. The main steps are:

- Develop course plans for your senior year and for one graduate year with your academic adviser and obtain Director of Graduate Studies (DGS) approval.
- Take the GRE exam in the senior fall or earlier
- Apply for admission to Duke's Graduate School (for a Master of Science) or Pratt's Master's Program Office (for a Master of Engineering or Master of Engineering Management program) during senior fall.

Advising

Assignment of ECE advisors: Students are assigned an ECE faculty advisor at the time they declare their major. Students can declare majors or change them at any time after their first year. To do this, the students complete an online Pratt Declaration of Major form.

The assignment of faculty advisors is based upon the major interests (e.g., computer engineering, integrated circuits, CompSci second major) expressed by the students and the need to balance the number of advisees per faculty.

First-year advising: In order to provide program information to first-year students interested in ECE, all students are invited to an orientation presented by the ECE Director of Undergraduate Studies (DUS) in mid-Fall. The presentation covers the degree requirements and commonly asked questions. Incoming students can also meet with their assigned academic advisor during the first few weeks of the semester. Students will meet with their academic advisor in October/November and March/April, during the pre-registration periods.

Pre-registration advising of ECE students: Students meet with their advisors during the pre-registration period in March/April for the Fall semester and in October/November for the Spring semester. The advisor reviews the student's academic report, discusses with the student the courses that he or she will take in the following semester, and makes the student 'eligible to enroll' in classes for the following semester. Should you make any changes to your course plan after meeting with your advisor it is your responsibility to inform your advisor of those changes. Your advisor is likely to keep notes on your meetings and this will keep his or her notes accurate and, more importantly, your change in plan may trigger a comment from your advisor that could be beneficial to you.

During the advising meeting you should discuss any concerns or problems that you might be having academically, and you are encouraged to initiate conversations about the field of electrical and computer engineering that you may be interested in exploring further, and/or career options within the discipline.

Career advising: Students can discuss their career plans with their advisors. In addition, Duke's Career Center is available for career advising, support for career exploration, and assistance with job search skills such as networking, writing professional documents, and interviewing in preparation for securing summer internships, full-time employment, and other professional opportunities.

Information on Internships and Employment

Information on internship and employment opportunities is posted on the website of the Duke University Career Center. In addition to maintaining the websites mentioned above, the Career Center organizes various career-related activities. These include (a) career advice sessions, (b) career skills workshops, (c) providing information about specific industries and opportunities, (d) alumni connections, and (d) workshops and seminars on internship and employment that are specific for engineering or ECE students. The workshops and seminars are announced via emails and posted on the plasma screens in the engineering buildings.

In addition to the Career Center, Kirsten Shaw (kirsten.shaw@duke.edu), Director of Corporate and Industry Relations, help Pratt students connect with corporations for internship opportunities. For more information, see the Undergraduate Internship Opportunities webpage.

Information on internships and employment is sometimes sent directly to the ECE faculty or the department. When this happens, the information will be distributed to ECE students via emails from the DUS Assistant.

Appendix A: Sample Programs of Study for the ECE Major and Select Second Majors, Minors, and Special Programs for Students Matriculating Fall 2021 through Summer 2025

Table A.1a

Electrical and Computer Engineering (ECE) Major (F2021<=matric<=Su2025)

First Year	
Fall Semester	**Spring Semester**
EGR 101L Engineering Design & Communication	ECE 110L Fundamentals of ECE
Math 111L Introductory Calculus I	Math 112L Introductory Calculus II
EGR 103L Computational Methods in Engineering or EGR 105L Computing for Engineers or CS 201 Data Structures & Algorithms[1]	Physics 151L Introductory Mechanics[3]
Writing 101 / Chem 101DL Core Concepts in Chemistry[2]	Chem 101DL Core Concepts in Chemistry / Writing 101
Sophomore Year	
Fall Semester	**Spring Semester**
ECE 280L Signals and Systems	ECE 230L / ECE 250D / ECE 270DL
COMPSCI 201 Data Structures and Algorithms (if not already taken to replace EGR 103L or EGR 105L)	ECE 230L / ECE 250D / ECE 270DL
Math 218D-2 Matrices and Vectors	Math 219 Multivariable Calculus
Physics 152L Intro Electricity, Magnetism, Optics[3]	Social Science or Humanities Elective 2
Social Science or Humanities Elective 1	
Junior Year	
Fall Semester	**Spring Semester**
ECE 230L / ECE 250D / ECE 270DL	ECE Elective[5]
ECE Area Elective 1[4]	ECE Area Elective 2
Math 353 Ordinary & Partial Differential Equations	Statistics Elective[6]
Social Science or Humanities Elective 3	Social Science or Humanities Elective 4
Senior Year	
Fall Semester	**Spring Semester**
ECE Extension Elective[7] or ECE Design Elective[8]	ECE Design Elective or ECE Extension Elective
ECE Area Elective 3	ECE Area Elective 4
Social Science or Humanities Elective 5	Free Elective
Free Elective	Free Elective
Free Elective	

1. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
2. AP credit for Chem 20 or 21 is also acceptable.
3. See also the Physics requirements.
4. ECE Area Electives: Four courses selected from the set approved for the ECE program. Courses must be selected from at least two areas, and at least two courses must be from the same area. See **Appendix D: Approved ECE Curricular Area Elective** for a complete course listing.
5. ECE Elective: Any ECE course at the 300 level or above.
6. Statistics Elective selected from the list of approved Statistics electives, found in Appendix E.
7. ECE Extension Elective: A 300-level course from the departments of Bio, Chem, CompSci, EvAnth, Math, Physics, Psych & Neuro, or Stats that has an NS or QS code (and does not have an SS code); or a 300-level course from ECE, BME, MEMS, or CEE; or DUS-approved 200-level course (found in Appendix E). **AP credits and non-ECE Indep Study courses cannot be used to satisfy this requirement.** See ECE Extension Elective for complete description.
8. Approved ECE Design Elective: Approved Electrical & Computer Engineering Design Elective taken after meeting all Math, Science, and ECE Core curriculum requirements. In addition, each approved design elective has one or more pre-requisite upper-level ECE courses. The elected design course may not simultaneously also count as an ECE Area Elective, ECE Extension Elective, or ECE Elective. See Appendix E for a list of all currently approved Design courses.

Table A.1b

ECE Major *with Concentration in Machine Learning* (F2021<=matric<=Su2025)

First Year	
Fall Semester	**Spring Semester**
EGR 101L Engineering Design & Communication	ECE 110L Fundamentals of ECE
Math 111L Introductory Calculus I	Math 112L Introductory Calculus II
EGR 103L Computational Methods in Engineering or EGR 105L Computing for Engineers or CS 201 Data Structures & Algorithms[1]	Physics 151L Introductory Mechanics[3]
Writing 101 / Chem 101DL Core Concepts in Chemistry[2]	Chem 101DL Core Concepts in Chemistry / Writing 101
Sophomore Year	
Fall Semester	**Spring Semester**
ECE 280L Signals and Systems	ECE 230L / ECE 250D / ECE 270DL
COMPSCI 201 Data Structures and Algorithms (if not already taken to replace EGR 103L or EGR 105L)	Statistics Elective[4]
Math 218D-2 Matrices and Vectors	Math 219 Multivariable Calculus
Physics 152L Intro Electricity, Magnetism, Optics[3]	Social Science or Humanities Elective 2
Social Science or Humanities Elective 1	
Junior Year	
Fall Semester	**Spring Semester**
ECE 230L / ECE 250D / ECE 270DL	ECE 230L / ECE 250D / ECE 270DL
ECE 480 Applied Prob. for Statistical Learning (ECE Area Elective/ML1)	ECE 580 Introduction to Machine Learning (ECE Elective/ML2)
Math 353 Ordinary & Partial Differential Equations	ECE Area Elective 4[5]
Social Science or Humanities Elective 3	Social Science or Humanities Elective 4
Senior Year	
Fall Semester	**Spring Semester**
ECE Extension Elective/ML5[6] or ECE Design Elective[7]	ECE Design Elective or ECE Extension Elective/ML5
ECE Area Elec 3/ML3[8] or ECE Elective/ML4[9]	ECE Elective/ML4 or ECE Area Elec 3/ML3
Social Science or Humanities Elective 5	Free Elective
Free Elective	Free Elective
Free Elective	

1. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
2. AP credit for Chem 20 or 21 is also acceptable.
3. See also the Physics requirements.
4. Statistics Elective selected from the list of approved Statistics electives, found in Appendix E.
5. ECE Area Elective 4: To ensure breadth, course must be selected from an area such that at least one of Area Elective 3 and Area Elective 4 is not from SPC&C area. See **Appendix D: Approved ECE Curricular Area Elective** for a complete course listing.
6. ECE Extension Elective/ML5: See ML Concentration Requirement #5
7. Approved ECE Design Elective: Approved Electrical & Computer Engineering Design Elective taken after meeting all Math, Science, and ECE Core curriculum requirements. In addition, each approved design elective has one or more pre-requisite upper-level ECE courses. The elected design course may not simultaneously also count as an ECE Area Elective, ECE Extension Elective, or ECE Elective. See Appendix E for a list of all currently approved Design courses.
8. ECE Area Elective 3/ML3: See ML Concentration Requirement #3
9. ECE Elective/ML4: See ML Concentration Requirement #4

Table A.2a

ECE with Computer Science (COMPSCI) Second Major (F2021<=matric<=Su2025)

NOTE: Students are advised to consult with CompSci regarding COMPSCI major requirements.

First Year	
Fall Semester	**Spring Semester**
EGR 101L Engineering Design & Communication	ECE 110L Fundamentals of ECE
Math 111L Introductory Calculus I	Math 112L Introductory Calculus II
EGR 103L Computational Methods in Engineering or EGR 105L Computing for Engineers or CS 201 Data Structures & Algorithms[1]	Physics 151L Introductory Mechanics[3]
Writing 101 / Chem 101DL Core Concepts in Chemistry[2]	Chem 101DL Core Concepts in Chemistry / Writing 101
Sophomore Year	
Fall Semester	**Spring Semester**
ECE 280L Signals and Systems	ECE 230L / ECE 270DL
COMPSCI 201 Data Structures and Algorithms (if not already taken to replace EGR 103L or EGR 105L)	ECE 250D Computer Architecture
Math 218D-2 Matrices and Vectors	Math 219 Multivariable Calculus
Physics 152L Intro Electricity, Magnetism, Optics[3]	Social Science or Humanities Elective 2
Social Science or Humanities Elective 1	
Junior Year	
Fall Semester	**Spring Semester**
ECE 230L / ECE 270DL	ECE Area Elective 1[6]
COMPSCI Systems Elective[4] or COMPSCI Elective[5]	COMPSCI Elective or COMPSCI Systems Elective
Math 353 Ordinary & Partial Differential Equations	Statistics Elective[7]
Social Science or Humanities Elective 3	Social Science or Humanities Elective 4
Senior Year	
Fall Semester	**Spring Semester**
ECE Area Elective 2 or ECE Elective[8]	ECE Elective or ECE Concentration Elective 2
ECE Area Elective 3 or ECE Design Elective[9]	ECE Design Elective or ECE Area Elective 3
COMPSCI 330 Design & Analysis of Algorithms[10]	COMPSCI Elective
ECE Area Elective 4 or COMPSCI Elective	COMPSCI Elective or ECE Area Elective 4
Social Science or Humanities Elective 5	

1. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
2. AP credit for Chem 20 or 21 is also acceptable.
3. See also the Physics requirements.
4. COMPSCI Systems Elective: Selected from one of the approved systems elective courses (consult with CompSci for current list).
5. COMPSCI Elective: Any COMPSCI elective at the 200 level or above. May not be an Independent Study course. If a CompSci course is crosslisted in ECE, the course may satisfy both a COMPSCI elective requirement and an ECE Area elective (if the ECE course is approved as one) or ECE Elective (if 300-level or higher). The COMPSCI elective then becomes a Free Elective. Enrollment should be under the ECE course number.
6. ECE Area Electives: Four courses selected from the set approved for the ECE program. Courses must be selected from at least two areas, and at least two courses must be from the same area. See **Appendix D: Approved ECE Curricular Area Elective** for a complete course listing.
7. Statistics Elective selected from the list of approved Statistics electives, found in Appendix E.
8. ECE Elective: Any ECE course at the 300 level or above.
9. Approved ECE Design Elective: Approved ECE Design Elective taken after meeting all Math, Science, and ECE Core curriculum requirements. In addition, each approved design elective has one or more pre-requisite upper-level ECE courses. The elected design course may not simultaneously also count as an ECE Area Elective, ECE Extension Elective or ECE Elective. See Appendix E for a list of all currently approved Design courses.
10. COMPSCI 330 is required of the CompSci major and satisfies the ECE Extension Elective requirement.

Table A.2b

ECE *with Concentration in ML* and COMPSCI Second Major (F2021<=matric<=Su2025)

NOTE: Students are advised to consult with CompSci regarding COMPSCI major requirements.

First Year	
Fall Semester	**Spring Semester**
EGR 101L Engineering Design & Communication	ECE 110L Fundamentals of ECE
Math 111L Introductory Calculus I	Math 112L Introductory Calculus II
EGR 103L Comp Methods in Eng or EGR 105L Computing for Engineers or CS 201 Data Structures & Algorithms[1]	Physics 151L Introductory Mechanics[3]
Writing 101 / Chem 101DL Core Concepts in Chemistry[2]	Chem 101DL Core Concepts in Chemistry / Writing 101
Sophomore Year	
Fall Semester	**Spring Semester**
ECE 280L Signals and Systems	ECE 250D Computer Architecture
COMPSCI 201 Data Structures and Algorithms (if not already taken to replace EGR 103L or EGR 105L)	Statistics Elective[4]
Math 218D-2 Matrices and Vectors	Math 219 Multivariable Calculus
Physics 152L Intro Electricity, Magnetism, Optics[3]	Social Science or Humanities Elective 2
Social Science or Humanities Elective 1	
Junior Year	
Fall Semester	**Spring Semester**
ECE 230L / ECE 270DL	ECE 230L / ECE 270DL
ECE 480 Applied Prob. for Statistical Learning (ECE Area Elective/ML1)	ECE 580 Introduction to Machine Learning (ECE Elective/ML2)
COMPSCI Systems Elective[5] or COMPSCI Elective[6]	COMPSCI Elective or COMPSCI Systems Elective
Math 353 Ordinary & Partial Differential Equations	Social Science or Humanities Elective 4
Social Science or Humanities Elective 3	
Senior Year	
Fall Semester	**Spring Semester**
ECE Extension Elective/ML5[7] or ECE Design Elective[8]	ECE Design Elective or ECE Extension Elective/ML5
ECE Area Elec 3/ML3[9] or ECE Elective/ML4[10]	ECE Elective/ML4 or ECE Area Elec 3/ML3
ECE Area Elective 4[11] or COMPSCI Elective	COMPSCI Elective or ECE Area Elective 4
COMPSCI 330 Design & Analysis of Algorithms	COMPSCI Elective
Social Science or Humanities Elective 5	

1. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
2. AP credit for Chem 20 or 21 is also acceptable.
3. See also the Physics requirements.
4. Statistics Elective selected from the list of approved Statistics electives, found in Appendix E.
5. COMPSCI Systems Elective: Selected from one of the approved systems elective courses (consult with CompSci for current list).
6. COMPSCI Elective: Any COMPSCI elective at the 200 level or above. May not be an Independent Study course. If a CompSci course is crosslisted in ECE, the course may satisfy both a COMPSCI elective requirement and an ECE Area elective (if the ECE course is approved as one) or ECE Elective (if 300-level or higher). The COMPSCI elective then becomes a Free Elective. Enrollment should be under the ECE course number.
7. ECE Extension Elective/ML5: See ML Concentration Requirement #5
8. Approved ECE Design Elective: Approved Electrical & Computer Engineering Design Elective taken after meeting all Math, Science, and ECE Core curriculum requirements. In addition, each approved design elective has one or more pre-requisite upper-level ECE courses. The elected design course may not simultaneously also count as an ECE Area Elective, ECE Extension Elective, or ECE Elective. See Appendix E for a list of all currently approved Design courses.
9. ECE Area Elective 3/ML3: See ML Concentration Requirement #3
10. ECE Elective/ML4: See ML Concentration Requirement #4
11. ECE Area Elective 4: To ensure breadth, course must be selected from an area such that at least one of Area Elective 3 and Area Elective 4 is not from SPC&C area. See **Appendix D: Approved ECE Curricular Area Elective** for a complete course listing.

Table A.3

ECE with BME Second Major (F2021<=matric<=Su2025)

First Year	
Fall Semester	**Spring Semester**
EGR 101L Engineering Design & Communication	Biology 201L Molecular Biology or, with Bio AP=5, Bio 203L
Chem 101DL Core Concepts in Chemistry or Chem 110DL or Chem 21 or Academic Writing 101	Chem 101DL Core Concepts in Chemistry or Chem 110DL or Chem 21 or Academic Writing 101
Math 111L Introductory Calculus I	Math 112L Introductory Calculus II
EGR 103L Computational Methods in Engineering or EGR 105L Computing for Engineers or CS 201 Data Structures & Algorithms[1]	Physics 151L Introductory Mechanics[2]
Sophomore Year	
Fall Semester	**Spring Semester**
ECE 110L Fundamentals of ECE	ECE 230L / ECE 270DL / ECE 280L
EGR 201L Mechanics of Solids	ECE 230L / ECE 270DL / ECE 280L
Math 218D-2 Matrices and Vectors	Math 219 Multivariable Calculus
Physics 152L Intro Electricity, Magnetism, Optics[2]	COMPSCI 201 Data Structures and Algorithms (if not already taken to replace EGR 103L or EGR 105L)
BME 244L Quantitative Physiology	Chem 210DL Modern Apps Chem Principles **or** Chem 201DL Organic Chemistry
Junior Year	
Fall Semester	**Spring Semester**
ECE 250D Computer Architecture	ECE Area Elective 1[3]
ECE 230L / ECE 270DL / ECE 280L	BME 354L Intro to Medical Instrumentation
Math 353 Ordinary & Partial Differential Equations	BME 301L Bioelectricity **or** BME 303 Modern Diagnostic Imaging Systems[4]
BME 221L Biomaterials or ME 221L Structure and Properties of Solids	Statistics Elective[5]
BME 260L Modeling Cell & Molecular Systems	Social Science or Humanities Elective 1
Senior Year	
Fall Semester	**Spring Semester**
ECE Elective[6] or ECE Area Elective 2	ECE Elective or ECE Area Elective 2
BME design: BME 436L or 464L or 474L[7]	BME General Elective
BME Advanced Elective[4]	Social Science or Humanities Elective 4
Social Science or Humanities Elective 2	Social Science or Humanities Elective 5
Social Science or Humanities Elective 3	

1. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
2. See also the Physics requirements.
3. ECE Area Electives: Two courses selected from the set approved for the ECE program. Both courses must be selected from the same area. See **Appendix D: Approved ECE Curricular Area Elective** for a complete course listing.
4. If BME 301L is taken, the Advanced Elective must be in the Electrobiology (EL) Area of Focus.
   If BME 303 is taken, the Advanced Elective must be in the Imaging and Measurements (IM) Area of Focus.
   See the BME Undergraduate Program Handbook for details.
5. Statistics Elective selected from the list of approved Statistics electives, found in Appendix E.
6. ECE Elective: Any ECE course at the 300 level or above.
7. BME 474L is the second course in a two-semester design sequence; BME 473L is prereq for it and must also be taken.

Table A.4

ECE with Physics Second Major (F2021<=matric<=Su2025)

First Year	
Fall Semester	**Spring Semester**
EGR 101L Engineering Design & Communication	ECE 110L Fundamentals of ECE
Math 111L Introductory Calculus I	Math 112L Introductory Calculus II
EGR 103L Computational Methods in Engineering or EGR 105L Computing for Engineers or CS 201 Data Structures & Algorithms[1]	Chem 101DL Core Concepts in Chemistry / Writing 101
Writing 101 / Chem 101DL Core Concepts in Chemistry[2]	Physics 151L Introductory Mechanics
	Physics 164L Introductory Experimental Physics I
Sophomore Year	
Fall Semester	**Spring Semester**
ECE 280L Signals and Systems	ECE 230L / ECE 250D / ECE 270DL
COMPSCI 201 Data Structures and Algorithms (if not already taken to replace EGR 103L or EGR 105L)	ECE 230L / ECE 250D / ECE 270DL
Math 218D-2 Matrices and Vectors	Math 219 Multivariable Calculus
Physics 152L Intro Electricity, Magnetism, Optics	Physics 361 Intermediate Mechanics
Physics 165L Introductory Experimental Physics II	Social Science or Humanities Elective 1
Junior Year	
Fall Semester	**Spring Semester**
ECE 230L / ECE 250D / ECE 270DL	ECE Area Elective 1 / Physics Elective 1[3]
Math 353 Ordinary & Partial Differential Equations	Statistics Elective[4]
Physics 264L Optics and Modern Physics	ECE Area Elective 2
Physics 362 Electricity & Magnetism	Physics 363 Thermal Physics[5]
Social Science or Humanities Elective 2	Social Science or Humanities Elective 3
Senior Year	
Fall Semester	**Spring Semester**
Approved ECE Design Elective[6] / Social Science or Humanities Elective 4	Social Science or Humanities Elective 4 / Approved ECE Design Elective
ECE Area Elective 3	ECE Area Elective 4 / Physics Elective 2[3]
Physics 464 Quantum Mechanics I	ECE Elective[7]
Physics 417S Advanced Physics Lab & Seminar	Social Science or Humanities Elective 5

1. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
2. AP credit for Chem 20 or 21 is also acceptable.
3. If two ECE Area Electives are taken from ECE 340L, 523, 621, or 623 (EP area), then these courses will also satisfy two Physics Electives (because they are crosslisted). If different ECE courses are chosen for these two ECE Area Electives, then additional Physics elective courses (one 200+, one 300+-level) must be taken to fulfill the Physics requirements. Of the remaining ECE Area Electives, at least one must be from an area other than Engineering Physics in order to satisfy the breadth requirement. See **Appendix D: Approved ECE Curricular Area Elective** for a complete course listing.
4. Statistics Elective selected from the list of approved Statistics electives, found in Appendix E.
5. Physics 363 satisfies the ECE Extension Elective requirement.
6. Approved ECE Design Elective: Approved Electrical & Computer Engineering Design Elective taken after meeting all Math, Science, and ECE Core curriculum requirements. In addition, each approved design elective has one or more pre-requisite upper-level ECE courses. The elected design course may not simultaneously also count as an ECE Area Elective, ECE Extension Elective, or ECE Elective. See Appendix E for a list of all currently approved Design courses
7. ECE Elective: Any ECE course at the 300-level or above.

Table A.5a

ECE Major w/ Energy Engineering Minor (ME 331L as Ext. Elec)
(F2021<=matric<=Su2025)

First Year	
Fall Semester	**Spring Semester**
EGR 101L Engineering Design & Communication	ECE 110L Fundamentals of ECE
Math 111L Introductory Calculus I	Math 112L Introductory Calculus II
EGR 103L Computational Methods in Engineering or EGR 105L Computing for Engineers or CS 201 Data Structures & Algorithms[1]	Physics 151L Introductory Mechanics[3]
Writing 101 / Chem 101DL Core Concepts in Chemistry[2]	Chem 101DL Core Concepts in Chemistry / Writing 101
Sophomore Year	
Fall Semester	**Spring Semester**
ECE 280L Signals and Systems	ECE 230L
COMPSCI 201 Data Structures and Algorithms (if not already taken to replace EGR 103L or EGR 105L)	ECE 250D / ECE 270DL
Math 218D-2 Matrices and Vectors	Math 219 Multivariable Calculus
Physics 152L Intro Electricity, Magnetism, Optics[3]	Social Science or Humanities Elective 2
Social Science or Humanities Elective 1	
Junior Year	
Fall Semester	**Spring Semester**
ECE 250D / ECE 270DL	ECE Elective[6]
ECE Area Elective 1[4]	ECE Area Elective 2
Math 353 Ordinary & Partial Differential Equations	Statistics Elective[7]
ECE Extension Elective[5] [ME 331L Thermodynamics (ENRGYEGR Elective 1, C&E area)]	ENRGYEGR Elective 2
Social Science or Humanities Elective 3	Social Science or Humanities Elective 4
Senior Year	
Fall Semester	**Spring Semester**
ECE Area Elective 3 or ECE Design Elective[8]	ECE Design Elective[8] or ECE Area Elective 3
ME 461 Energy Engineering and the Environment (ENRGYEGR Core)	ECE Area Elective 4
ENRGYEGR Elective 3	ENRGYEGR Elective 4
Bass Connections 395 (0.5 cr., prereq for EGR 424L)	EGR 424L (ENRGYEGR capstone)
Social Science or Humanities Elective 5	

1. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
2. AP credit for Chem 20 or 21 is also acceptable.
3. See also the Physics requirements.
4. ECE Area Electives: Four courses selected from the set approved for the ECE program. Courses must be selected from at least two areas, and at least two courses must be from the same area. See **Appendix D: Approved ECE Curricular Area Electives** for a complete course listing.
5. ECE Extension Elective: ME 331L may be used to satisfy the ECE Extension Elective requirement.
6. ECE Elective: Any ECE course at the 300 level or above.
7. Statistics Elective selected from the list of approved Statistics electives, found in Appendix E.
8. Approved ECE Design Elective: Approved Electrical & Computer Engineering Design Elective taken after meeting all Math, Science, and ECE Core curriculum requirements. In addition, each approved design elective has one or more pre-requisite upper-level ECE courses. The elected design course may not simultaneously also count as an ECE Area Elective, ECE Extension Elective, or ECE Elective. See Appendix E for a list of all currently approved Design courses.

Table A.5b

ECE Major with Energy Engineering Minor (ECE Design as capstone) (F2021<=matric<=Su2025)

First Year	
Fall Semester	**Spring Semester**
EGR 101L Engineering Design & Communication	ECE 110L Fundamentals of ECE
Math 111L Introductory Calculus I	Math 112L Introductory Calculus II
EGR 103L Computational Methods in Engineering or EGR 105L Computing for Engineers or CS 201 Data Structures & Algorithms[1]	Physics 151L Introductory Mechanics[3]
Writing 101 / Chem 101DL Core Concepts in Chemistry[2]	Chem 101DL Core Concepts in Chemistry / Writing 101
Sophomore Year	
Fall Semester	**Spring Semester**
ECE 280L Signals and Systems	ECE 230L
COMPSCI 201 Data Structures and Algorithms (if not already taken to replace EGR 103L or EGR 105L)	ECE 250D / ECE 270DL
Math 218D-2 Matrices and Vectors	Math 219 Multivariable Calculus
Physics 152L Intro Electricity, Magnetism, Optics[3]	Social Science or Humanities Elective 2
Social Science or Humanities Elective 1	
Junior Year	
Fall Semester	**Spring Semester**
ECE 250D / ECE 270DL	ECE Elective[5]
ECE Area Elective 1[4]	ECE Area Elective 2
Math 353 Ordinary & Partial Differential Equations	Statistics Elective[6]
Social Science or Humanities Elective 3	Social Science or Humanities Elective 4
ME 331L Thermodynamics (ENRGYEGR Elec 1, C&E)	ENRGYEGR Elective 2
Senior Year	
Fall Semester	**Spring Semester**
ECE Area Elective 3	ECE Design Elective (also ENRGYEGR capstone)[7]
ME 461 Energy Engineering and the Environment (ENRGYEGR Core)	ECE Area Elective 4
ENRGYEGR Elective 3	ECE Extension Elective[8]
Social Science or Humanities Elective 5	ENRGYEGR Elective 4

1. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
2. AP credit for Chem 20 or 21 is also acceptable.
3. See also the Physics requirements.
4. ECE Area Electives: Four courses selected from the set approved for the ECE program. Courses must be selected from at least two areas, and at least two courses must be from the same area. See **Appendix D: Approved ECE Curricular Area Electives** for a complete course listing.
5. ECE Elective: Any ECE course at the 300 level or above.
6. Statistics Elective selected from the list of approved Statistics electives, found in Appendix E.
7. Approved ECE Design Elective: Approved Electrical & Computer Engineering Design Elective taken after meeting all Math, Science, and ECE Core curriculum requirements. In addition, each approved design elective has one or more pre-requisite upper-level ECE courses. The elected design course may not simultaneously also count as an ECE Area Elective, ECE Extension Elective, or ECE Elective. See Appendix E for a list of all currently approved Design courses. The ECE Design course may also count as the ENERGYEGR capstone course with approval by the ENERGYEGR Minor program director.
8. ECE Extension Elective: A 300-level course from the departments of Bio, Chem, CompSci, EvAnth, Math, Physics, Psych & Neuro, or Stats that has an NS or QS code (and does not have an SS code); or a 300-level course from ECE, BME, MEMS, or CEE; or DUS-approved 200-level course (found in Appendix E). **AP credits and non-ECE Indep Study courses cannot be used to satisfy this requirement.** See ECE Extension Elective for complete description

Table A.6a

ECE Major with Junior Fall semester at the Marine Lab (F2021<=matric<=Su2025)

First Year	
Fall Semester	**Spring Semester**
EGR 101L Engineering Design & Communication	ECE 110L Fundamentals of ECE
Math 111L Introductory Calculus I	Math 112L Introductory Calculus II
EGR 103L Computational Methods in Engineering or EGR 105L Computing for Engineers or CS 201 Data Structures & Algorithms[1]	Physics 151L Introductory Mechanics[3]
Writing 101 / Chem 101DL Core Concepts in Chemistry[2]	Chem 101DL Core Concepts in Chemistry / Writing 101
Sophomore Year	
Fall Semester	**Spring Semester**
ECE 280L Signals and Systems	ECE 230L / ECE 250D / ECE 270DL
COMPSCI 201 Data Structures and Algorithms (if not already taken to replace EGR 103L or EGR 105L)	ECE 230L / ECE 250D / ECE 270DL
Math 218D-2 Matrices and Vectors	Math 219 Multivariable Calculus
Physics 152L Intro Electricity, Magnetism, Optics[3]	Social Science or Humanities Elective 2
Social Science or Humanities Elective 1	
Junior Year	
Fall Semester (at the Marine Lab)	**Spring Semester**
ECE Elective[4] [ECE 391 Independent Study]	ECE 230L / ECE 250D / ECE 270DL
ECE Extension Elective[5]	ECE Area Elective 1[6]
Math 353 Ordinary & Partial Differential Equations	Statistics Elective[7]
Social Science or Humanities Elective 3	Social Science or Humanities Elective 4
Senior Year	
Fall Semester	**Spring Semester**
Approved ECE Design Elective[8] / ECE Area Elective 2	ECE Area Elective 2 / Approved ECE Design Elective
ECE Area Elective 3	ECE Area Elective 4
Social Science or Humanities Elective 5	Free Elective
Free Elective	Free Elective
Free Elective	

1. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
2. AP credit for Chem 20 or 21 is also acceptable.
3. See also the Physics requirements.
4. ECE Elective: Any ECE course at the 300 level or above.
5. ECE Extension Elective: A 300-level course from the departments of Bio, Chem, CompSci, EvAnth, Math, Physics, Psych & Neuro, or Stats that has an NS or QS code (and does not have an SS code); or a 300-level course from ECE, BME, MEMS, or CEE; or DUS-approved 200-level course (found in Appendix E). **AP credits and non-ECE Indep Study courses cannot be used to satisfy this requirement.** See ECE Extension Elective for complete description.
6. ECE Area Electives: Four courses selected from the set approved for the ECE program. Courses must be selected from at least two areas, and at least two courses must be from the same area. See **Appendix D: Approved ECE Curricular Area Elective** for a complete course listing.
7. Statistics Elective selected from the list of approved Statistics electives, found in Appendix E.
8. Approved ECE Design Elective: Approved Electrical & Computer Engineering Design Elective taken after meeting all Math, Science, and ECE Core curriculum requirements. In addition, each approved design elective has one or more pre-requisite upper-level ECE courses. The elected design course may not simultaneously also count as an ECE Area Elective, ECE Extension Elective, or ECE Elective. See Appendix E for a list of all currently approved Design courses.

Table A.6b

ECE Major with Junior Spring semester at the Marine Lab (F2021<=matric<=Su2025)

First Year	
Fall Semester	**Spring Semester**
EGR 101L Engineering Design & Communication	ECE 110L Fundamentals of ECE
Math 111L Introductory Calculus I	Math 112L Introductory Calculus II
EGR 103L Computational Methods in Engineering or EGR 105L Computing for Engineers or CS 201 Data Structures & Algorithms[1]	Physics 151L Introductory Mechanics[3]
Writing 101 / Chem 101DL Core Concepts in Chemistry[2]	Chem 101DL Core Concepts in Chemistry / Writing 101
Sophomore Year	
Fall Semester	**Spring Semester**
ECE 280L Signals and Systems	ECE 230L / ECE 250D / ECE 270DL
COMPSCI 201 Data Structures and Algorithms (if not already taken to replace EGR 103L or EGR 105L)	ECE 230L / ECE 250D / ECE 270DL
Math 218D-2 Matrices and Vectors	Math 219 Multivariable Calculus
Physics 152L Intro Electricity, Magnetism, Optics[3]	Social Science or Humanities Elective 2
Social Science or Humanities Elective 1	
Junior Year	
Fall Semester	**Spring Semester (at the Marine Lab)**
ECE 230L / ECE 250D / ECE 270DL	ECE 461A Ocean Engineering[4,5]
ECE Area Elective 1[6]	ECE 366A/ENV 335A Drones in Marine Science[4,5]
Math 353 Ordinary & Partial Differential Equations	Free Elective (e.g. ECE 384LA Marine Bioacoustics)
Social Science or Humanities Elective 3	Social Science or Humanities Elective 4 (e.g. MARSCI 571A Urban Tropical Ecology, Travel course, Singapore)
Statistics Elective[7]	
Senior Year	
Fall Semester	**Spring Semester**
Approved ECE Design Elective[8] / ECE Area Elective 2	ECE Area Elective 2 / Approved ECE Design Elective
ECE Area Elective 3	ECE Area Elective 4
Social Science or Humanities Elective 5	Free Elective
Free Elective	Free Elective

1. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
2. AP credit for Chem 20 or 21 is also acceptable.
3. See also the Physics requirements.
4. ECE Elective: Any ECE course at the 300 level or above.
5. ECE Extension Elective: A 300-level course from the departments of Bio, Chem, CompSci, EvAnth, Math, Physics, Psych & Neuro, or Stats that has an NS or QS code (and does not have an SS code); or a 300-level course from ECE, BME, MEMS, or CEE; or DUS-approved 200-level course (found in Appendix E). **AP credits and non-ECE Indep Study courses cannot be used to satisfy this requirement.** See ECE Extension Elective for complete description.
6. ECE Area Electives: Four courses selected from the set approved for the ECE program. Courses must be selected from at least two areas, and at least two courses must be from the same area. See **Appendix D: Approved ECE Curricular Area Elective** for a complete course listing.
7. Statistics Elective selected from the list of approved Statistics electives, found in Appendix E.
8. Approved ECE Design Elective: Approved Electrical & Computer Engineering Design Elective taken after meeting all Math, Science, and ECE Core curriculum requirements. In addition, each approved design elective has one or more pre-requisite upper-level ECE courses. The elected design course may not simultaneously also count as an ECE Area Elective, ECE Extension Elective, or ECE Elective. See Appendix E for a list of all currently approved Design courses.

Appendix B: Check Sheets for the ECE Major and Select Second Majors

Electrical & Computer Engineering
ECE Major Requirements (2021<=matric<=F2025)

_____
Name

		Semester Taken	Grade	Notes
	Writing			
1.	Writing 101			
	Mathematics and Natural Science			
2.	MATH 111L [or AP credit for Math 21]			
3.	MATH 112L [or AP credit for Math 22]			
4.	MATH 218D-2			
5.	MATH 219			
6.	MATH 353			
7.	Statistics elective _____			Approved options: Stat 240L or Math 230, 231, 340 or EGR 238L or ECE 380 or ECE 555
8.	CHEM 101DL [or AP credit for Chem 20 or 21]			
9.	PHY 151L [or AP credit for Physics 25]			
10.	PHY 152L [or AP credit for Physics 26 + approved PHY course][1]			
	ECE Foundation Courses			
11.	EGR 101L			
12.	EGR 103L or EGR 105L[2]			
13.	COMPSCI 201			
14.	ECE 110L			
15.	ECE 230L			
16.	ECE 250D			
17.	ECE 270DL			
18.	ECE 280L			
	ECE Concentration[3] & Application Electives			
19.	ECE Area Elective 1 _____			
20.	ECE Area Elective 2 _____			
21.	ECE Area Elective 3 _____			
22.	ECE Area Elective 4 _____			
23.	ECE Extension Elective[4] _____			
24.	ECE Elective _____			Any ECE course at the 300-level or above
25.	ECE Design Elective _____			ECE 449, 458, 459, 469, 487, 489, 532, 539, or 559

	Humanities and Social Sciences[5]		**Area**	
26.	_____			
27.	_____			
28.	_____			
29.	_____			
30.	_____			

	Electives			
31.	_____			
32.	_____			
33.	_____			
34.	_____			

1. Students must take (or have AP credit for) PHY 151L and 152L, and at least one physics course must be taken post-matriculation. If a student has AP credit for both 25 & 26, then one of PHY 153L, 163D, 264, 361, or 362 must be taken. Alternatively, AP credit for 26 can be waived by taking 152L.
2. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
3. Courses must be selected from at least two areas, and at least two courses must be from the same area. The same course may not be used as a Area Elective and count as a required Design Elective, ECE Extension Elective, or ECE Elective.
4. A 300-level course from the departments of Bio, Chem, CompSci, EvAnth, Math, Physics, Psych & Neuro, or Stats that has an NS or QS code (and does not have an SS code), or a 300-level course from ECE, BME, MEMS, CEE, or DUS-approved 200-level course (see App. E). AP credit and non-ECE Independent Study cannot be used to satisfy ECE Extension Elective requirement.
5. Five required. At least one SS and 2 from remaining areas of knowledge (FL, CZ, ALP). For depth, two must be from the same approved department or program – and one course must be at the 200-level or above. See Pratt Policies & Procedures for detailed list of approved departments and programs (http://www.pratt.duke.edu/undergrad/policies/3505#humanities).

Electrical & Computer Engineering

ECE Major *with Concentration in ML* (2021<=matric<=F2025)

Name _____

		Semester Taken	Grade	Notes
	Writing			
1.	Writing 101			
	Mathematics and Natural Science			
2.	MATH 111L [or AP credit for Math 21]			
3.	MATH 112L [or AP credit for Math 22]			
4.	MATH 218D-2			
5.	MATH 219			
6.	MATH 353			
7.	Statistics elective _____			Approved options: Stat 240L or Math 230, 231, 340 or EGR 238L or ECE 380 or ECE 555
8.	CHEM 101DL [or AP credit for Chem 20 or 21]			
9.	PHY 151L [or AP credit for Physics 25]			
10.	PHY 152L [or AP credit for Physics 26 + approved PHY course][1]			
	ECE Foundation Courses			
11.	EGR 101L			
12.	EGR 103L or EGR 105L[2]			
13.	COMPSCI 201			
14.	ECE 110L			
15.	ECE 230L			
16.	ECE 250D			
17.	ECE 270DL			
18.	ECE 280L			
	ECE Concentration & Application Electives			
19.	ECE Area Elective 1/ML1 ECE 480			
20.	ECE Area Elective 2/ML2 ECE 580			
21.	ECE Area Elective 3/ML3 _____			ECE 588, 661, 662, 663, 682D, 684, 685D, 687D, or ECE 590 Area Elective on ML (w/ approval)
22.	ECE Area Elective 4[3] _____			
23.	ECE Extension Elective/ML5 _____			ECE 588, 661, 662, 663, 682D, 684, 685D, 687D, 689, or ECE 495/496/590 Special Topic w/ ML focus (w/ approval); CS 371, 527; Math 412, 465; ME 555, Stat 340, 360.
24.	ECE Elective/ML4 _____			ECE 588, 661, 662, 663, 682D, 684, 685D, 687D, 689, or 495/496/590 Special Topics course on ML (w/ approval).
25.	ECE Design Elective _____			ECE 449, 458, 459, 469, 487, 489, 532, 539, or 559

	Humanities and Social Sciences[4]		Area	
26.	_____			
27.	_____			
28.	_____			
29.	_____			
30.	_____			

	Electives		
31.	_____		
32.	_____		
33.	_____		
34.	_____		

1. Students must take (or have AP credit for) PHY 151L and 152L, and at least one physics course must be taken post-matriculation. If a student has AP credit for both 25 & 26, then one of PHY 153L, 163D, 264, 361, or 362 must be taken. Alternatively, AP credit for 26 can be waived by taking 152L.
2. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
3. To ensure breadth, course must be selected from an area of concentration such that at least one of CE#3 and CE#4 is not from SPC&C area.
4. Five required. At least one SS and 2 from remaining areas of knowledge (FL, CZ, ALP). For depth, two must be from the same approved department or program – and one course must be at the 200-level or above. See Pratt Policies & Procedures for detailed list of approved departments and programs (http://www.pratt.duke.edu/undergrad/policies/3505#humanities).

Electrical & Computer Engineering

ECE with COMPSCI (BS) Major (2021<=matric<=F2025)

Name _____

NOTE: Students are advised to consult with CS regarding COMPSCI requirements.

		Semester Taken	Grade	Notes
	Writing			
1.	Writing 101			
	Mathematics and Natural Science			
2.	MATH 111L [or AP credit for Math 21]			
3.	MATH 112L [or AP credit for Math 22]			
4.	MATH 218D-2			
5.	MATH 219			
6.	MATH 353			
7.	Statistics elective _____			Approved options: Stat 240L or Math 230, 231, 340 or EGR 238L or ECE 380 or ECE 555
8.	CHEM 101DL [or AP credit for Chem 20 or 21]			
9.	PHY 151L [or AP credit for Physics 25]			
10.	PHY 152L [or AP credit for Physics 26 + approved PHY course][1]			
	ECE Foundation Courses			
11.	EGR 101L			
12.	EGR 103L or EGR 105L[2]			
13.	COMPSCI 201			
14.	ECE 110L			
15.	ECE 230L			
16.	ECE 250D			
17.	ECE 270DL			
18.	ECE 280L			
	ECE Concentration[3] & Application Electives			
19.	ECE Area Elective 1 _____			
20.	ECE Area Elective 2 _____			
21.	ECE Area Elective 3 _____			
22.	ECE Area Elective 4 _____			
23.	ECE Extension Elective[4] COMPSCI 330			
24.	ECE Elective _____			Any ECE course at the 300-level or above
25.	ECE Design Elective _____			ECE 449, 458, 459, 469, 487, 489, 532, 539, or 559

	Humanities and Social Sciences[5]		**Area**	
26.	_____			
27.	_____			
28.	_____			
29.	_____			
30.	_____			

	Electives			
31.	COMPSCI Systems Elective[6] _____			COMPSCI 310 (or 510), 316 (or 516), 345, 350L (or 550), 351 (or 581), 356 (or 514), or 512
32.	COMPSCI Elective[6] _____			200-level or higher, may not be an Indep. Study
33.	COMPSCI Elective[6] _____			200-level or higher, may not be an Indep. Study
34.	COMPSCI Elective[6] _____			200-level or higher, may not be an Indep. Study

1. Students must take (or have AP credit for) PHY 151L and 152L, and at least one physics course must be taken post-matriculation. If a student has AP credit for both 25 & 26, then one of PHY 153L, 163D, 264, 361, or 362 must be taken. Alternatively, AP credit for 26 can be waived by taking 152L.
2. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
3. Courses must be selected from at least two areas, and at least two courses must be from the same area. The same course may not be used as a Area Elective and count as a required Design Elective, ECE Extension Elective, or ECE Elective.
4. CompSci 330, which is required for the CompSci major, fulfills the ECE Extension Elective requirement.
5. Five required. At least one SS and 2 from remaining areas of knowledge (FL, CZ, ALP). For depth, two must be from the same approved department or program – and one course must be at the 200-level or above. See Pratt Policies & Procedures for detailed list of approved departments and programs (http://www.pratt.duke.edu/undergrad/policies/3505#humanities).
6. If a CompSci course is crosslisted in ECE, the course may satisfy both a CompSci elective requirement and an ECE Area Elective (if the ECE course is approved as one) or ECE Elective (if 300-level or higher). The CompSci elective then becomes a Free Elective.

Electrical & Computer Engineering

ECE *with Conc in ML* and COMPSCI Major (2021<=matric<=F2025)

Name _____

NOTE: Students are advised to consult with CS regarding COMPSCI requirements.

		Semester Taken	Grade	Notes
	Writing			
1.	Writing 101			
	Mathematics and Natural Science			
2.	MATH 111L [or AP credit for Math 21]			
3.	MATH 112L [or AP credit for Math 22]			
4.	MATH 218D-2			
5.	MATH 219			
6.	MATH 353			
7.	Statistics elective _____			Approved options: Stat 240L or Math 230, 231, 340 or EGR 238L or ECE 380 or ECE 555
8.	CHEM 101DL [or AP credit for Chem 20 or 21]			
9.	PHY 151L [or AP credit for Physics 25]			
10.	PHY 152L [or AP credit for Physics 26 + approved PHY course][1]			
	ECE Foundation Courses			
11.	EGR 101L			
12.	EGR 103L or EGR 105L[2]			
13.	COMPSCI 201			
14.	ECE 110L			
15.	ECE 230L			
16.	ECE 250D			
17.	ECE 270DL			
18.	ECE 280L			
	ECE Concentration & Application Electives			
19.	ECE Area Elective 1/ML1 ECE 480			
20.	ECE Area Elective 2/ML2 ECE 580			
21.	ECE Area Elective 3/ML3 _____			ECE 588, 661, 662, 663, 682D, 684, 685D, 687D, or ECE 590 Area Elective on ML (w/ approval)
22.	ECE Area Elective 4[3] _____			
23.	ECE Extension Elective/ML5 _____			ECE 588, 661, 662, 663, 682D, 684, 685D, 687D, 689, or ECE 495/496/590 Special Topic w/ ML focus (w/ approval); CS 371, 527; Math 412, 465; ME 555, Stat 340, 360.
24.	ECE Elective/ML4 _____			ECE 588, 661, 662, 663, 682D, 684, 685D, 687D, 689, or 495/496/590 Special Topics course on ML (w/ approval).
25.	ECE Design Elective _____			ECE 449, 458, 459, 469, 487, 489, 532, 539, or 559

	Humanities and Social Sciences[4]		Area	
26.	_____			
27.	_____			
28.	_____			
29.	_____			
30.	_____			

	Electives			
31.	COMPSCI 330 _____			
32.	COMPSCI Systems Elective[5] _____			COMPSCI 310 (or 510), 316 (or 516), 345, 350L (or 550), 351 (or 581), 356 (or 514), or 512
33.	COMPSCI Elective[5] _____			200-level of higher, may not be an Indep. Study
34.	COMPSCI Elective[5] _____			200-level of higher, may not be an Indep. Study
35.	COMPSCI Elective[5] _____			200-level of higher, may not be an Indep. Study

1. Students must take (or have AP credit for) PHY 151L and 152L, and at least one physics course must be taken post-matriculation. If a student has AP credit for both 25 & 26, then one of PHY 153L, 163D, 264, 361, or 362 must be taken. Alternatively, AP credit for 26 can be waived by taking 152L.
2. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
3. To ensure breadth, course must be selected from an area of concentration such that at least one of CE#3 and CE#4 is not from SPC&C area.
4. Five required. At least one SS and 2 from remaining areas of knowledge (FL, CZ, ALP). For depth, two must be from the same approved department or program – and one course must be at the 200-level or above. See Pratt Policies & Procedures for detailed list of approved departments and programs (http://www.pratt.duke.edu/undergrad/policies/3505#humanities).
5. If a CompSci course is crosslisted in ECE, the course may satisfy both a COMPSCI elective requirement and an ECE Area Elective (if the ECE course is approved as one) or ECE Elective (if 300-level or higher). The COMPSCI elective then becomes a Free Elective.

Electrical & Computer Engineering

ECE with BME Second Major (2021<=matric<=F2025)

Name _____

		Semester Taken	Grade	Notes
	Writing			
1.	Writing 101			
	Mathematics and Natural Science			
2.	MATH 111L [or AP credit for Math 21]			
3.	MATH 112L [or AP credit for Math 22]			
4.	MATH 218D-2			
5.	MATH 219			
6.	MATH 353			
7.	Statistics elective _____			Approved options: Stat 240L or Math 230, 231, 340 or EGR 238L or ECE 380 or ECE 555
8.	CHEM 101DL [AP credit for Chem 21]			Alternate option : CHEM 110DL
9.	CHEM 201DL or CHEM 210DL			
10.	PHY 151L [or AP credit for Physics 25]			
11.	PHY 152L [or AP credit for Physics 26 + approved PHY course][1]			
12.	Biology 201L or [if Bio AP=5, Bio 203L]			
	ECE and BME Courses			
13.	EGR 101L			
14.	EGR 103L or EGR 105L[2]			
15.	COMPSCI 201			
16.	ECE 110L			
17.	ECE 230L			
18.	ECE 250D			
19.	ECE 270DL			
20.	ECE 280L			
21.	ECE Area Elective 1[3] _____			
22.	ECE Area Elective 2[3] _____			
23.	ECE Elective _____			Any ECE course at the 300-level or above
24.	BME 244L			
25.	EGR 201L			
26.	BME 221L or ME 221L			
27.	BME 260L			
28.	BME 354L			
29.	BME 301L or BME 303L[4]			
30.	BME Advanced Elective [4] _____			
31.	BME General Elective _____			
32.	BME Design Elective _____			Approved options: BME 436L or 464L or 474L

	Humanities and Social Sciences[5]		**Area**	
33.	_____			
34.	_____			
35.	_____			
36.	_____			
37.	_____			

1. Students must take (or have AP credit for) PHY 151L and 152L, and at least one physics course must be taken post-matriculation. If a student has AP credit for both 25 & 26, then one of PHY 153L, 163D, 264, 361, or 362 must be taken. Alternatively, AP credit for 26 can be waived by taking 152L.
2. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
3. Courses must be selected from the same concentration area.
4. If BME 301L is taken the Advanced Elective must be from the Electrobiology (EL) Area
   If BME 303L is taken the Advanced Elective must be from the Imaging and Measurement Systems (IM) Area
5. Five required. At least one SS and 2 from remaining areas of knowledge (FL, CZ, ALP). For depth, two must be from the same approved department or program – and one course must be at the 200-level or above. See Pratt Policies & Procedures for detailed list of approved departments and programs (http://www.pratt.duke.edu/undergrad/policies/3505#humanities).

Electrical & Computer Engineering
ECE with PHYSICS Second Major (2021<=matric<=F2025)

Name _____

		Semester Taken	Grade	Notes
	Writing			
1.	Writing 101			
	Mathematics and Natural Science			
2.	MATH 111L [or AP credit for Math 21]			
3.	MATH 112L [or AP credit for Math 22]			
4.	MATH 218D-2			
5.	MATH 219			
6.	MATH 353			
7.	Statistics elective _____			Approved options: Stat 240L or Math 230, 231, 340 or EGR 238L or ECE 380 or ECE 555
8.	CHEM 101DL [or AP credit for Chem 20 or 21]			
9.	PHY 151L [or AP credit for Physics 25]			
10.	PHY 152L [or if AP credit for Physics 25 & 26, then PHY 163D][1]			
11a.	PHY 164L (0.5 credit)			
11b.	PHY 165L (0.5 credit)			
12.	PHY 264L			
13.	PHY 361			
14.	PHY 362			
15.	PHY 363			Satisfies the ECE Extension elective
16.	PHY 417S			
17.	PHY 464			
	ECE Foundation Courses			
18.	EGR 101L			
19.	EGR 103L or EGR 105L[2]			
20.	COMPSCI 201			
21.	ECE 110L			
22.	ECE 230L			
23.	ECE 250D			
24.	ECE 270DL			
25.	ECE 280L			
	ECE Concentration & Application Electives			
26.	ECE Conc Elec 1 (EP) / PHY Elec 1[3] _____			One of: ECE 340L, 523, 621, or 623
27.	ECE Conc Elec 2 (EP) / PHY Elec 2[3] _____			One of: ECE 340L, 523, 621, or 623
28.	ECE Area Elective 3 (non-EP) _____			
29.	ECE Area Elective 4 (any area) _____			
30.	ECE Elective _____			Any ECE course at the 300-level or above
31.	ECE Design Elective _____			ECE 449, 458, 459, 469, 487, 489, 532, 539, or 559

	Humanities and Social Sciences[4]		**Area**	
32.	_____			
33.	_____			
34.	_____			
35.	_____			
36.	_____			

1. The sequence of PHY 161L and 162L or PHY 161D/164L and PHY 162D/165L is also accepted in lieu of PHY 151L and 152L. Students with AP credit for PHY 25 and 26 are encouraged to take PHY 163D in lieu of the 151L/152L, 161L/162L, or 161D/164L/162D/165L sequences.
2. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
3. Two of ECE 340L (PHY 320L), ECE 523 (PHY 627), ECE 621 (PHY 628), and ECE 623 (PHY 623) can be used to satisfy ECE Area Electives 1 & 2 (EP area) and also satisfy 2 PHY electives since they are crosslisted. If different ECE courses are chosen for ECE Conc Elec 1 & ECE Conc Elec 2, then additional Physics elective courses (one 200+, one 300+-level) must be taken to fulfill the Physics requirements.
4. Five required. At least one SS and 2 from remaining areas of knowledge (FL, CZ, ALP). For depth, two must be from the same approved department or program – and one course must be at the 200-level or above. See Pratt Policies & Procedures for detailed list of approved departments and programs (http://www.pratt.duke.edu/undergrad/policies/3505#humanities).

Electrical & Computer Engineering

ECE Major *with Energy Engineering Minor* (2021<=matric<=F2025)

Name _____

		Semester Taken	Grade	Notes
	Writing			
1.	Writing 101			
	Mathematics and Natural Science			
2.	MATH 111L [or AP credit for Math 21]			
3.	MATH 112L [or AP credit for Math 22]			
4.	MATH 218D-2			
5.	MATH 219			
6.	MATH 353			
7.	Statistics elective _____			Approved options: Stat 240L or Math 230, 231, 340 or EGR 238L or ECE 380 or ECE 555
8.	CHEM 101DL [or AP credit for Chem 20 or 21]			
9.	PHY 151L [or AP credit for Physics 25]			
10.	PHY 152L [or AP credit for Physics 26 + approved PHY course][1]			
	ECE Foundation Courses			
11.	EGR 101L			
12.	EGR 103L or EGR 105L[2]			
13.	COMPSCI 201			
14.	ECE 110L			
15.	ECE 230L			
16.	ECE 250D			
17.	ECE 270DL			
18.	ECE 280L			
	ECE Concentration[3] & Application Electives			
19.	ECE Area Elective 1			
20.	ECE Area Elective 2			
21.	ECE Area Elective 3 _____			
22.	ECE Area Elective 4 _____			
23.	ECE Extension Elective ME 331L			Also satisfies ENRGYEGR Elective 1
24.	ECE Elective _____			Any ECE course at the 300-level or above
25.	ECE Design Elective _____			ECE 449, 458, 459, 469, 487, 489, 532, 539, or 559

	Humanities and Social Sciences[4]		**Area**	
26.	_____			
27.	_____			
28.	_____			
29.	_____			
30.	_____			

	Electives			
31.	ENRGYEGR Elective 2			
32.	ENRGYEGR Elective 3			
33.	ENRGYEGR Elective 4			
34.	ME 461			ENRGYEGR Core
35.	Bass Connections 395 (0.5 credit)			Prerequisite for ENRGYEGR capstone
36.	EGR 424L			ENRGYEGR capstone

1. Students must take (or have AP credit for) PHY 151L and 152L, and at least one physics course must be taken post-matriculation. If a student has AP credit for both 25 & 26, then one of PHY 153L, 163D, 264, 361, or 362 must be taken. Alternatively, AP credit for 26 can be waived by taking 152L.
2. Students who place into CompSci 201 are not required to take EGR 103L or EGR 105L.
3. Courses must be selected from at least two areas, and at least two courses must be from the same area. The same course may not be used as a Area Elective and count as a required Design Elective, ECE Extension Elective, or ECE Elective.
4. Five required. At least one SS and 2 from remaining areas of knowledge (FL, CZ, ALP). For depth, two must be from the same approved department or program – and one course must be at the 200-level or above. See Pratt Policies & Procedures for detailed list of approved departments and programs (http://www.pratt.duke.edu/undergrad/policies/3505#humanities).

Appendix C: Major-specific Requirements for the Minor in ECE

The basic requirements for a minor in ECE (described in detail on p. 13) include three courses at the foundational/core level and two upper-level courses. Below are major-specific modifications (e.g., courses that are disallowed for the Minor in ECE because students are required to taken essentially equivalent courses for their major).

Majors in Pratt

ME Major/ECE Minor: Path #1

If an ME major has taken EGR224L, then

- ***Core courses (choose at least one, and up to three)***[1]:
  - *ECE 230L Microelectronic Devices & Circuits*,
  - *ECE 250D Computer Architecture*,
  - *ECE 270DL Fields & Waves*.
- ***Upper-level courses***: Take a minimum of two upper-level courses[2]. Student may choose to replace up to two (of three) ECE core courses with additional upper-level ECE courses to meet the minimum requirement of 5 ECE courses.

[1] An ME major cannot take ECE 110L or ECE 280L, however, EGR 224L will satisfy prerequisites in lieu of ECE 110L and ECE 280L.

[2] Because the ME major requires courses essentially equivalent to ECE 110L and ECE 280L, a student majoring in ME can choose to reduce the number of ECE core courses taken to fulfill Minor requirements to as few as one, and take additional upper-level courses to meet the minimum requirement of 5 ECE courses.

ME Major/ECE Minor: Path #2

If an ME major has taken ECE 110L, then

- Student should take ECE 280L in lieu of EGR 224L to satisfy ME major requirement, but ECE 280L will not count toward the ECE minor.
- ***Core courses (choose at least two, and up to three)***:
  - *ECE 110L Fundamentals of ECE*
  - *ECE 230L Microelectronic Devices & Circuits*,
  - *ECE 250D Computer Architecture*,
  - *ECE 270DL Fields & Waves*.
- ***Upper-level courses***: Take a minimum of two upper-level courses[1]. Student may choose to replace one (of three) ECE core courses with an additional upper-level ECE course to meet the minimum requirement of 5 ECE courses.

[1] Because the ME major requires course essentially equivalent to ECE 280L, a student majoring in ME can choose to reduce the number of ECE core courses taken to fulfill Minor requirements to as few as two, and take additional upper-level courses to meet the minimum requirement of 5 ECE courses.

BME Major/ECE Minor

- ***Core courses (choose at least one, and up to three)***[1,2]:
  - *ECE 230L Microelectronic Devices & Circuits*,
  - *ECE 250D Computer Architecture*,
  - *ECE 270DL Fields & Waves*.
- ***Upper-level courses***: Take a minimum of two upper-level courses[3]. Student may choose to replace up to two (of three) ECE core courses with additional upper-level ECE courses to meet the minimum requirement of 5 ECE courses.

[1] ECE 110L cannot count toward the ECE Minor (ECE 110L is required for BME major)
[2] ECE 280L cannot count toward the ECE Minor (BME 271 or ECE 280L is required for BME major)
[3] Because the BME major requires ECE 110L and ECE 280L or their equivalents, a student majoring in BME can choose to reduce the number of ECE core courses taken to fulfill Minor requirements to as few as one, and take additional upper-level courses to meet the minimum requirement of 5 ECE courses.

Majors in Trinity

- Physics 142L or Physics 162L are considered equivalent to Physics 152L for purposes of meeting ECE course prerequisites.

COMPSCI Major/ECE Minor

- ***Core courses (choose at least two, and up to three)***[1]:
  - *ECE 110L Fundamentals of ECE*,
  - *ECE 230L Microelectronic Devices & Circuits*,
  - *ECE 270DL Fields & Waves*,
  - *ECE 280L Signals & Systems*
- ***Upper-level courses***: Take a minimum of two upper-level courses[2]. Student may choose to replace up to one (of three) ECE core courses with additional upper-level ECE courses.

[1] ECE 250D cannot count toward the ECE Minor (COMPSCI 250D is equivalent)
[2] Because the COMPSCI major requires COMPSCI 210 or 250D, a student majoring in COMPSCI can choose to reduce the number of ECE core courses taken to fulfill Minor requirements to as few as two, and take additional upper-level courses to meet the minimum requirement of 5 ECE courses.

PHYSICS Major/ECE Minor

- ***Core courses (choose at least two, and up to three)***[1]:
  - *ECE 110L Fundamentals of ECE*,
  - *ECE 230L Microelectronic Devices & Circuits*,
  - *ECE 250D Computer Architecture,*
  - *ECE 280L Signals & Systems*
- ***Upper-level courses***: Take a minimum of two upper-level courses[2]. Student may choose to replace up to one (of three) ECE core courses with additional upper-level ECE courses.

[1] ECE 270DL cannot count toward the ECE Minor due to a substantially equivalent requirement for the Physics major.
[2] Because the PHYSICS major requires courses with significant overlap with ECE 270DL, a student majoring in PHYSICS can choose to reduce the number of ECE core courses taken to fulfill Minor requirements to as few as two, and take additional upper-level courses to meet the minimum requirement of 5 ECE courses.

Appendix D: Approved ECE Curricular Area Electives

Computer Engineering and Digital Systems

The discipline concerned with the operation and design of computers and computer-based systems. Although analog computers, in which electrical signals directly represent physical quantities, were historically important in the development of modern computers (and continue to be used in some systems), digital computers are predominant and are the primary focus of the computer engineering in the ECE department in the Pratt School. The Computer Engineering and Digital Systems curriculum begins with a core course in computer architecture (ECE 250D). Students can then study logic design (ECE 350L – the foundation course), computer networking, VLSI chip design, and other advanced topics. Computer engineering interfaces strongly with many other areas of electrical engineering (electronics, electromagnetics, signal processing, and control theory) as well as with computer science.

- ECE 350L Digital Systems [ECE 250D & COMPSCI 201]
- ECE 351 Software Engineering [COMPSCI 201] (ECE 495/496 version before Fall of 2025 also counts)
- ECE 353 Intro to Operating Systems (C/L COMPSCI 310) [ECE 250D or COMPSCI 250D]
- ECE 356 Computer Network Architecture [ECE 250D or COMPSCI 250D]
- ECE 459 Introduction to Embedded System [ECE 330L or ECE 331L or ECE 350L]
- ECE 536 Synthesis & Verification of VLSI Systems [ECE 350L]
- ECE 538 VLSI System Testing [ECE 350L]
- ECE 539 Full Custom VLSI Design [ECE 350L and ECE 331L]
- ECE 552 Advanced Computer Architecture I [ECE 250D or COMPSCI 250D]
- ECE 553 Compiler Construction [ECE 250D or COMPSCI 250D]
- ECE 554 Fault-Tolerant and Testable Computer Systems [ECE 250D or COMPSCI 250D]
- ECE 555 Probability for Electrical and Computer Engineers [MATH 216]
- ECE 556 Wireless Networking and Mobile Computing [ECE 356 or COMPSCI 310]
- ECE 557 Perform. and Reliability of Computer Networks [ECE 356 and ECE 555]
- ECE 558 Computer Networks and Distributed Systems (C/L COMPSCI 514)
- ECE 559 Advanced Digital System Design [co-req ECE 331L and pre-req 350L]
- ECE 560 Computer & Information Security [COMPSCI 310/ECE 353]
- ECE 564 Mobile Application Development [COMPSCI 307D or COMPSCI 308 or ECE 651]
- ECE 565 Performance Optimization and Parallelism [ECE 250D and COMPSCI 310 and ECE 552]
- ECE 566 Enterprise Storage Architecture [COMPSCI 310/ECE 353]
- ECE 567 Cyberphysical System Design [ECE 350L and ECE 353/COMPSCI 310]
- ECE 590 Cyberphysical System Formal Methods (S17) [ECE 350L and ECE 353/COMPSCI 310]
- ECE 590 Emerging Memory and Computer Architecture (F17, S19) [ECE 250D]
- ECE 590 Conventional & Emerging Memory Systems (S20)
- ECE 652 Advanced Computer Architecture II [ECE 552 or COMPSCI 550]
- ECE 654 Edge Computing [ECE/CS 350L or ECE/CS 356 or CS 310/ECE 353]
- ECE 655 Full-Stack IoT Systems [ECE 250D or COMPSCI 250D]
- ECE 657 Usable Security and Privacy [Recommended: COMPSCI 201]
- ECE 661 Computer Engineering Machine Learning and Deep Neural Networks [CS 201]
- ECE 663 Machine Learning in Adversarial Settings [ECE 580 or ECE 687D or CS 371]

Signal Processing, Communications, and Control Systems

The disciplines concerned with representing, storing, interpreting, and transmitting information in systems of finite capacity in the presence of interference and noise; with extracting information from speech, image, video, radar, sonar, and medical data signals; and with using information, including feedback information comparing actual and desired system states, for controlling, shaping and stabilizing system performance in the presence of noise, delay, and inertia. Applications include telecommunications, intra- and inter-system communications, remote sensing, imaging, robotics, feed-back stabilized electronics, and the remote control of electro-mechanical systems, both large and small.

- ECE 381 Fundamentals of Digital Signal Processing [ECE 280L]
- ECE 382L Linear Control Systems [ECE 280L]
- ECE 383 Introduction to Robotics and Automation [ECE 280L or ME 224L]
- ECE 480 Applied Probability for Statistical Learning [Math 216 or Math 218D-2, and statistics]
- ECE 483 Introduction to Digital Communication Systems [ECE 280L and statistics]
- ECE 485 Digital Audio and Acoustic Signal Processing [ECE 280L and statistics]
- ECE 488 Digital Image and Multidimensional Signal [ECE 280L and statistics]
- ECE 495 Intro to Natural Language Processing (F24) [Math 218D-2 and (EGR 103L or CS 201) and statistics]
- ECE 580 Introduction to Machine Learning [(Math 216 or Math 218D-2) and CS 201 and ECE 480]
- ECE 588 Image and Video Processing [ECE 280L and (Math 216 or Math 218D-2) and statistics]
- ECE 590 The Fourier Transform and Applications (S23, S25) [ECE 280L or ECE 270DL]
- ECE 661 Computer Engineering Machine Learning and Deep Neural Networks [CS 201]
- ECE 681 Pattern Class & Recognition Tech [ECE 280L and (Math 216 or 218D-2) and statistics]
- ECE 682D Probabilistic Machine Learning [(Math 216 or Math 218D-2) and (Stat 250 or Stat 611)]
- ECE 684 Natural Language Processing [ECE 480 or ECE 580 or ECE 581 or ECE 682D]
- ECE 685D Introduction to Deep Learning [ECE 580 or ECE 681 or ECE 682D/STA 561D/CS 571D]
- ECE 687D Theory & Algorithms for Machine Learning [ECE 580 or ECE 681 or ECE 682D/STA 561D/CS 571D]

Solid-State Devices and Integrated Circuits

This area is concerned with the properties and manufacture of building-block devices (diodes, transistors, lasers) used in integrated circuits to build electronic and photonic systems. Example applications include: digital computer components (CPUs, RAM, GPUs), telecommunications equipment components (parts essential for cell phones, digital switches, wifi routers), and displays, which underlie a large array of consumer products (televisions, tablets, touch-screen systems). It also encompasses the burgeoning field of microelectromechanical, micromechanical and microfluidic devices made possible by the fabrication techniques underlying integrated circuit manufacture.

- ECE 330L Microelectronic Devices [ECE 230L]
- ECE 331L Fundamentals of Microelectronic Circuits [ECE 230L]
- ECE 431 Power Electronic Circuits for Energy Conversion [ECE 230L or EGR 224L]
- ECE 461/461A Ocean Engineering [ECE 230L or ECE 250D or ECE 270DL or ECE 280L]
- ECE 511 Foundations of Nanoscale Science & Technology [Physics 152L & Chem 101DL]
- ECE 512 Emerging Nanoelectronic Devices [ECE 230L]
- ECE 524 Introduction to Solid State Physics [ECE 521 or Phys 264L]
- ECE 526 Devices for Integrated Circuit [ECE 230L]
- ECE 527 Analog Integrated Circuits [ECE 526]
- ECE 528 Nanoscale Integrated Circuit Chip Technology [ECE 230L and Chem 101DL]
- ECE 529 Digital Integrated Circuits [ECE 331L]
- ECE 532 Analog Integrated Circuit Design [ECE 330L or ECE 331L]
- ECE 539 Full Custom VLSI Design [ECE 350L and ECE 331L]

Engineering Physics

The discipline concerned with the underlying laws of nature (specifically those governing electromagnetic, optical, and quantum phenomena) and how those laws can be applied. Application areas for electromagnetics, optics, and quantum mechanics are extremely broad and include communication systems, radar, radio astronomy, transmission lines, waveguides, optical fibers, cameras, remote sensing, quantum-based communication encryption, and quantum computing.

- ECE 340L Optics and Photonics [ECE 270DL]
- ECE 370D Intermediate Electromagnetic Theory [ECE 270DL]
- ECE 420/520 (Grad) Intro to Quantum Engineering [ECE 270DL and ECE 280L and Math 218D-2]
- ECE 521 Quantum Mechanics [Math 216 or Math 218D-2]
- ECE 522 Quantum Engineering with Atoms [ECE 521 or PHY 464] (course taught as ECE 590 S23 also counts)
- ECE 523 Quantum Computing [ECE 521 or PHYSICS 464]
- ECE 524 Introduction to Solid State Physics [ECE 521 or Phys 264L]
- ECE 541 Advanced Optics [ECE 340L]
- ECE 542 Holography and Coherent Imaging [ECE 270DL]
- ECE 543 Statistical Optics [ECE 270DL]
- ECE 571 Electromagnetic Theory [ECE 270DL]
- ECE 572 Electromagnetic Communication Systems [ECE 270DL]
- ECE 574 Waves in Matter [ECE 270DL]
- ECE 575 Microwave Electronic Circuits [ECE 270DL]
- ECE 577 Computational Electromagnetics [ECE 270DL]
- ECE 621 Quantum Error Correction [ECE 420 or ECE 520 or ECE 523]
- ECE 623 Quantum Information Theory [ECE 521 or PHYSICS 464 or (Math 216 or 218D-2)]

Photonics

The discipline concerned with the application of optical and optoelectronic technologies in information science. Photonic applications include information transmission on fiber and free space networks, data storage on disks and volume media, visible and infrared imaging systems, and displays. The Duke photonics program emphasizes hands-on experience with optical systems in communications, sensing, and display applications. Photonic engineering at Duke spans experiences as diverse as logical layer analysis of network protocols over fiber systems, analysis and testing of fiber dispersion, materials studies for optical memory, design of 3D microscopes for biomedical imaging, testing of liquid crystal materials and interfaces, analysis and construction of quantum dynamic systems, and explorations of laser-material and laser-tissue interactions.

- ECE 340L Optics & Photonics [ECE 270DL]
- ECE 341L Solar Cells [Physics 152L or equivalent]
- ECE 449 Sensors and Sensor Interface Design [ECE core courses and one of (ECE 330L or ECE 331L or ECE 340L or ECE 350L)]
- ECE 523 Quantum Computing [ECE 521 or PHYSICS 464]
- ECE 545 Foundations of Nanoelectronics and Nanophotonics [ECE 230L and ECE 270DL]
- ECE 546 Optoelectronic Devices for Optical Fiber Network [ECE 526]

Appendix E: Approved ECE Design, Natural Science, and Statistics Electives

Table E.1 Approved ECE Design Electives

ECE 449	Sensors and Sensor Interface Design *(prior to 2024, typically offered in the Spring; offered Fall 2024 and 2025 and not Spring 2024, 2025, or 2026)*
ECE 458	Engineering Software for Maintainability *(typically offered in the Spring)*
ECE 459	Embedded Systems Design *(typically offered in the Fall)*
ECE 469	Wearable & Ubiquitous Computing Systems Design *(typically offered in Spring)*
ECE 487	System Design for Machine Learning & Signal Processing *(typically offered Sp)*
ECE 532	Analog Integrated Circuit Design *(typically offered in the Spring)*
ECE 539	CMOS VLSI Design Methodologies *(typically offered in the Fall)*

Table E.2 Approved 200-level ECE Extension Electives

Biology 201L	Gateway to Biology: Molecular Biology
Biology 202L	Gateway to Biology: Genetics and Evolution
Biology 203L	Molecular Biology, Genetics, and Evolution
Biology 275A	Biology for Engineers: Informing Engineering Decisions
Chemistry 201DL	Organic Chemistry I
Chemistry 202DL	Organic Chemistry II
Chemistry 210DL	Modern Applications of Chemical Principles
Physics 264L	Optics and Modern Physics

AP credits may not be used to satisfy this requirement.

Table E.3 Approved ECE Statistics Courses

Statistics 240L	Probability for Statistical Inference, Modeling, and Data Analysis
Math 230	Probability
Math 231	Algorithmic Introduction to Probability
Math 340	Advanced Introduction to Probability
EGR 238L/Math 238L	Data and Decision Science
ECE 380	Introduction to Random Signals and Noise
ECE 555	Probability for Electrical and Computer Engineers

If ECE 555 is elected, it may not be double-counted as an ECE Area Elective or ECE Elective. If ECE 380 is elected, it may not be double-counted as an ECE Elective.